

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Civil Examination

May 2011

CL208 ADVANCED SURVEYING

Date: 10.5.2011, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1(a)** For trigonometric levelling having base of object is not accessible, Derive the formulae to find horizontal distance and elevation when the instrument axis at both the stations P & Q are at the same level. **03**
- (b)** Explain any four of following **04**
- a) Celestial horizon
 - b) Zenith distance
 - c) Vertical circle
 - d) Latitude of a line
 - e) Altitude
- Q - 2(a)** What is tacheometer? Write advantages of tacheometric surveying and Explain the procedure of finding out coefficients in the field. **06**
- (b)** A tacheometer was set up at a point P. Determine the gradient from point A to a point B from the following observations made with a tacheometer fitted with an analytic lens. The constant of tacheometer was 100 and staff held vertically. **06**

Staff	Reading	Vertical Angle	Staff Reading		
A	134°	10°32'	1.360	1.915	2.470
B	224°	5°6'	1.065	1.885	2.705

- (c)** With sight horizontal, staff held vertically 1.645 and 2.840 corresponding reading obtained as stadia reading. The distance between the focal length and object glass was 120m. The focal length of object glass is 20 cm. Distance between object glass and trunnion axis is 15 cm. Calculate stadia interval. **02**
- OR**
- Q - 2 (a)** Differentiate fixed hair method and movable hair method of tacheometry. **03**
- (b)** Discuss the subtense diaphragm method of tacheometric survey. **03**
- (c)** Explain tangential methods of surveying and derive the equations for finding out horizontal distances and elevation for both the angles are angles of depression. **03**
- (d)** Tacheometer is placed at a station A, reading on staff held on bench mark is 100.0 m and station B are 0.640, 2.200, 3.760 and 0.010, 2.120, 4.230 respectively. The angle of depression of telescope in 1st case -6°19' and of 2nd case is -7°42'. Using tacheometric constants as 100 & 0.3, Find out horizontal distance from A and B and reduced level of station B. **05**
- Q - 3 (a)** A theodolite was setup at a point P and staff was kept at a station Q. The distance PQ was 5000m. If the angle of elevation to a vane 4.0m above the foot of the staff was 5°. Determine the reduced level of top R, the elevation of instrument axis was 100.00 m. Also find reduced level of Q. **04**
- (b)** Discuss the coordinate systems to locate heavenly body on celestial sphere. **03**
- (c)** Write short note on **Attempt any two** **07**
- a) Opaque signals for geodetic survey.
 - b) Extension of base line
 - c) Primary triangulation system

OR

- Q - 3 (a) Determine the reduced level of top of television tower surrounded by transmission buildings and other administrative blocks. A theodolite was setup at a P in an adjoining park and angle of elevation was measured and found to be $33^{\circ}30'$. The reading on a staff held on a bench mark with telescope in horizontal plane was 1.002 m. The theodolite was transited and another point Q which is 100 m away was fixed. The angle of elevation of the top was measured at Q and found $26^{\circ}30'$. The reading on staff held on bench mark was again 1.002 m. If reduced level of bench mark is 200.005 m, Calculate the reduced level at top of television tower. 04
- (b) Explain the following angles in hours minute and seconds 03
- a) $78^{\circ}42'45''$ c) $25^{\circ}38'48''$
b) $10^{\circ}16'14''$
- (c) Write short note on following (Attempt any two) 07
- a) Requirement of good signals
b) Objectives of triangulation survey
c) Tertiary triangulation system

SECTION - II

- Q - 4 (a) What is the function of aerial camera? Describe schematically its essential parts. 07
- Q - 5 (a) Describe the principles of remote sensing. Explain actual remote sensing. 05
- (b) Find out the most probable value of angle A from the following equations 04
- $2A = 30^{\circ}30'31''$ wt.1
 $3A = 45^{\circ}45'48''$ wt.2
 $4A = 61^{\circ}1'00''$ wt.3
- (c) What are the different types of errors in surveying? Explain any one in detail. 05

OR

- Q - 5 (a) Explain theory of least square. 04
- (b) Explain the following terms 04
- a) Most probable value e) True error
b) Residual f) Most probable error
c) Standard deviation
- (c) Application of remote sensing in modern surveying. 06
- Q-6 (a) Discuss application of GIS in civil engineering 06
- (b) Write a short note on stereoscope and parallax bar 06
- (c) Explain laws of weight. 02

OR

- Q-6 (a) Write a short note on most probable values of quantities. 03
- (b) Attempt any three 09
- a) Global positioning system c) Passive remote sensing
b) Electromagnetic spectrum d) Visual interpretation
- (c) Discuss terrestrial photogrammetry and aerial photogrammetry. 02
